IMIPENEM + CILASTATIN

Seizure risk

Reserve antibiotic

CHEMICAL NAME Imipenem (N-formimidoyl thienamycin carbapenem) + cilastatin sodium (renal dehydropeptidase-I inhibitor)

CLASS Carbapenem + renal dipeptidase inhibitor

TRADE NAMES Primaxin · Tienam · Tenacid

THERAPEUTIC USES / INDICATIONS

- Severe multidrug-resistant Gram-negative infections (ESBL).
- Polymicrobial intra-abdominal and gynaecological infections.
- Hospital-acquired pneumonia, sepsis, febrile neutropenia.
- Bone, joint and complicated skin infections.

ADMINISTRATION

- **IV infusion:** ≤500 mg over 20–30 min; 1 g over 40–60 min.
- Slow the infusion if nausea occurs.
- Not for meningitis (seizure risk).

DOSAGES

- **Adult:** 500 mg every 6 h or 1 g every 6–8 h (dose as imipenem); max 4 g/day.
- **Children (>3 months):** 15–25 mg/kg every 6 h (max 2–4 g/day).
- Reduce dose in CrCl <90 mL/min per product table; avoid if CrCl <15 without dialysis.

PHARMACOKINETICS

- **Onset:** immediate IV.
- Imipenem is broken down by renal **dehydropeptidase-I**; cilastatin blocks this → active urinary levels and less nephrotoxicity.
- Excreted in urine. **t½:** 1 h.

PHARMACODYNAMICS

- Inhibits **cell-wall synthesis** → bactericidal.
- Very broad spectrum: Gram-positive (incl. *E. faecalis*), Gram-negative, *Pseudomonas*, anaerobes.

SIDE & ADVERSE EFFECTS

SIDE EFFECTS (COMMON)

- Nausea and vomiting (related to infusion rate).
- Diarrhoea, rash.
- Phlebitis.

ADVERSE EFFECTS (SERIOUS)

- Seizures (higher risk: CNS disease, renal impairment, high doses).
- Anaphylaxis.
- *C. difficile* colitis.
- Myoclonus, confusion.

CONTRAINDICATIONS & CAUTIONS

- Hypersensitivity to carbapenems or severe β-lactam allergy.
- **Caution:** CNS disorders, epilepsy, renal impairment, elderly.

KEY INTERACTIONS

- **Valproate:** ↓ valproate levels → seizures.
- **Ganciclovir:** ↑ seizures.
- **Ciclosporin:** ↑ CNS toxicity.

NURSING MANAGEMENT

- Infuse at correct rate; slow down if nausea.
- Monitor neurological status; seizure precautions in at-risk patients.
- Check renal function for dosing.
- Monitor temperature, cultures, CBC.
- Watch for diarrhoea.

PATIENT EDUCATION

- Report twitching, confusion, rash or diarrhoea.
- Tell staff about epilepsy medicines.

KEY POINTS

- Carbapenem + cilastatin (protects from renal breakdown).
- Slow infusion if nausea.
- Highest seizure risk among carbapenems.
- Avoid with valproate.